



**BIRLA INSTITUTE OF TECHNOLOGY AND SCIENCE, Pilani**  
**Pilani Campus**  
**AUGS/ AGSR Division**

**First SEMESTER 2026-27**

**Course Handout Part II**

Date: 01/08/2026

In addition to Part-I (General Handout for all courses appended to the timetable), this portion gives further specific details regarding the course.

Course No. : **ME F461**  
Course Title : **Refrigeration and Air-conditioning**  
Instructor-in-Charge : **MANOJ KUMAR SONI**

**Course Description:**

Principles, thermodynamic analysis, load estimates and design of various refrigeration and air conditioning systems for comfort and industrial applications. Theoretical or experimental investigation of refrigeration and air-conditioning problems.

**Scope and Objective of the Course:**

The course is designed to give a comprehensive understanding of the theoretical principles and practical aspects of the refrigeration and air-conditioning. The basic governing principles and refrigeration cycles, thermodynamic analysis and the associated components analysis and design of refrigeration systems will be covered. The comprehensive understanding of the design and implementation of the air-conditioning processes and systems will also be discussed.

**Textbooks:**

1. Arora, C.P. Refrigeration and Air-conditioning, 4th Ed., McGraw-Hill India, 2021.

**Reference books**

1. Stoecker, W. F., Jones, J. W. Refrigeration and Air Conditioning 2nd Ed.
2. Arora, R.C. Refrigeration and Air-conditioning, Prentice Hall India, 2010.
3. Faye C. McQuiston, Jerald D. Parker, Jeffrey D. Spitler, Heating, ventilating and air conditioning: analysis and design, 6th ed., Wiley, 2004.
4. ASHRAE Handbook

**Course Plan:**

| Lecture No. | Learning objectives       | Topics to be covered                                                                                                                                            | Chapter Textbook |
|-------------|---------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------|
| 1-2         | Introduction and review   | Introduction, history, and applications, Reversed Carnot Cycle                                                                                                  | 1-2              |
| 3-7         | Vapor Compression Systems | VCRS, Vapor compression systems calculations, Effect of operating condition on VCRS, Actual vapor compression systems, Cascade Systems, Multistage compression, | 3,5              |



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| Lecture No. | Learning objectives                                                  | Topics to be covered                                                                                                                                                                                              | Chapter Textbook |
|-------------|----------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------|
|             |                                                                      | Multi-Evaporator systems                                                                                                                                                                                          |                  |
| 8-11        | Compressors                                                          | Types of compressors, thermodynamic process, principle and performance characteristics of reciprocating compressors, rotary and centrifugal compressor                                                            | 6                |
| 12-13       | Condensers and Evaporators                                           | Types and heat transfer in condensers and evaporators, pressure drop in evaporators, Wilson's plot                                                                                                                | 7,9              |
| 14-15       | Expansion Devices                                                    | Types of expansion devices and design of capillary tubes and ejector expansion                                                                                                                                    | 8                |
| 16          | Complete system analysis                                             | Design and analysis of a complete vapor compression systems                                                                                                                                                       | 10               |
| 17-18       | Refrigerants                                                         | Designation, comparative study of refrigerants and their selection, chemical and physical requirements, and substitute of refrigerants                                                                            | 4                |
| 19-20       | Vapor Absorption Systems                                             | Principle of vapor absorption systems, Li-Br vapor absorption systems                                                                                                                                             | 12               |
| 21-23       | Air Cycle Refrigeration                                              | Limitation of Carnot cycle, reversed Brayton cycle, Aircraft refrigeration, The Joule-Thompson coefficient and inversion curve                                                                                    | 10               |
| 24          | HVAC Introduction                                                    | Introduction to heating ventilation and air-conditioning                                                                                                                                                          | Class notes      |
| 25-28       | Moist air properties and Psychrometry of air- conditioning processes | Introduction, Psychrometric properties, WBT, basic of air-conditioning processes, Summer and winter air-conditioning                                                                                              | 14-16            |
| 29-30       | Building Heat Transfer                                               | Fabric heat gain, periodic heat transfer, empirical methods for heat transfer evaluation                                                                                                                          | 18               |
| 31-34       | Heating and Cooling load calculation                                 | Heating and cooling load estimate, internal heat gain, system heat gains, requirement of energy conservation                                                                                                      | 19               |
| 35-36       | Solar radiation                                                      | Distribution of solar radiation, heat gain, shading                                                                                                                                                               | 17               |
| 37-38       | Transmission and distribution of air                                 | Friction and dynamic losses of ducts, air flow through simple ducts, transmission and distribution of air in rooms, duct design, requirements of a proper air distribution system, definition of air distribution | 21               |



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|-------------|---------------------|----------------------|------------------|
|             |                     | performance index    |                  |

**Evaluation Scheme:**

| Component                                        | Duration | Weightage (%) | Date & Time        | Nature of Component  |
|--------------------------------------------------|----------|---------------|--------------------|----------------------|
| Mid Semester Exam                                | 90       | 25            | TBA                | Closed Book          |
| Lecture tests (Surprise)                         |          | 10            | During class hours | Closed Book          |
| Assignments/ Projects/ Activity-based evaluation |          | 25            | Announced later    |                      |
| Comprehensive Exam                               | 180      | 40            | TBA                | Closed and Open Book |

**Reading assignments:** From time to time, reading assignments will be given to the students. These reading assignments are part of the course, and questions may appear in tests/examinations in these portions also.

**Chamber Consultation Hour:** TBA in the class

**Notices:** All notices concerning this course shall be displayed on the Nalanda only.

**Makeup Policy:** Make-up shall be given only to genuine cases and will be entertained through the AUGSD portal only. No makeup for the surprise components.

**Minimum Passing Marks Criteria:** *The passing marks for this course will be determined based on one of the two criteria: 20% of the median of the post-comprehensive examination score OR 20% of the average of the first three top scorers. The lower value between these two criteria will be considered as passing marks in this course. NC may also be awarded if a student misses regular evaluation components (except lecture surprise tests) even if he/she satisfies the above criteria.*

**Instructor-in-charge**  
ME F461 Refrigeration and Air-conditioning